

Problem 37. Find all pairs of nonnegative integers (x, y) satisfying

$$2^x = 5^y + 3.$$

Problem 38. Suppose that each positive integer is used once to label each of the vertices of an infinite graph and no vertices are left unlabelled. An edge exists between two vertices labelled by positive integers a and b if and only if

$$a + b + 1 \mid a^2 + b^2 + 1.$$

Is this graph connected?

Problem 39. Let I be the incentre of $\triangle ABC$ and J the excentre opposite vertex A . Denote by M the midpoint of BC and by N the midpoint of $\widehat{BAC}$. If T is the point symmetric to point N with respect to point A , prove that J, M, I, T are concyclic.

Problem 40. Let $\{x_n\}$ and $\{y_n\}$ be sequences of real numbers satisfying $x_1 = y_1 = 1$,

$$x_{n+1} = \frac{x_n + 2}{x_n + 1} \quad \text{and} \quad y_{n+1} = \frac{y_n^2 + 2}{2y_n} \quad \text{for } n = 1, 2, \dots$$

Prove that $y_{n+1} = x_{2^n}$ for $n = 0, 1, 2, \dots$